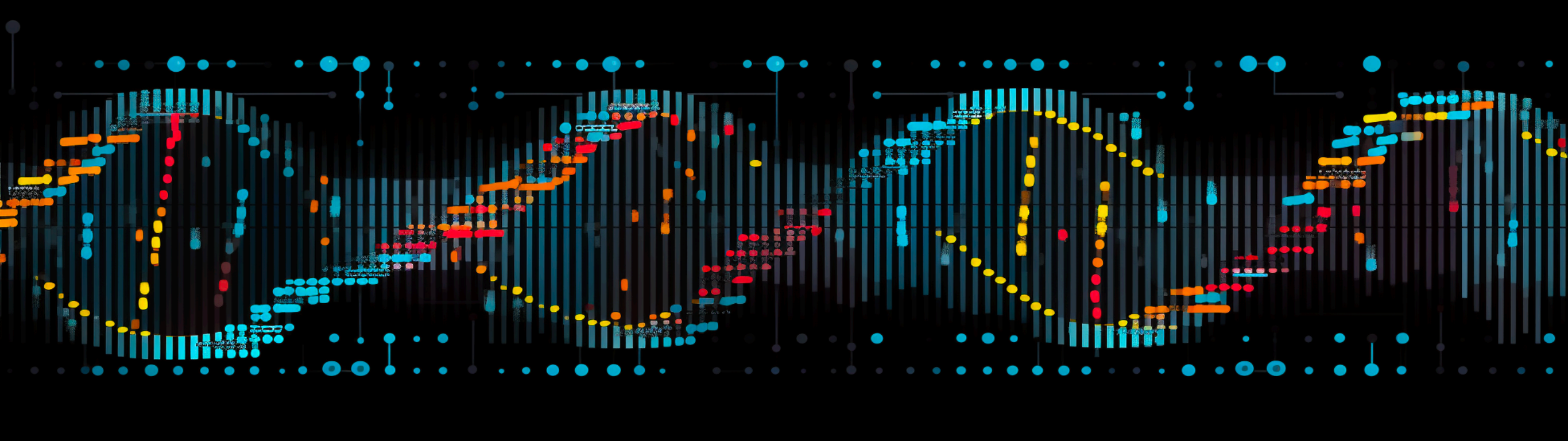




# Introduction to bioinformatics

## Chapter 4 Machine learning in bioinformatics - Network and clustering

Jiaxing Chen

# Outline

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- **Biological network**
- Network statistics
- Network models
  - Network generation models
  - Scale free
  - Small world

# Why networks?

- Networks provide natural description of relation between various components
- Examples:
  - Protein–protein interaction network
  - Protein domain co-occurrence network
  - Metabolic networks
  - Transcription Networks

# Biological Networks – Basic Components

- Nodes



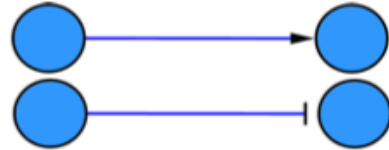
- DNA/RNA/Protein/Metabolite/Ontology

- Edges



Directed

- Distinction between source and target



- Activation (direct/indirect)

- Repression (direct/indirect)

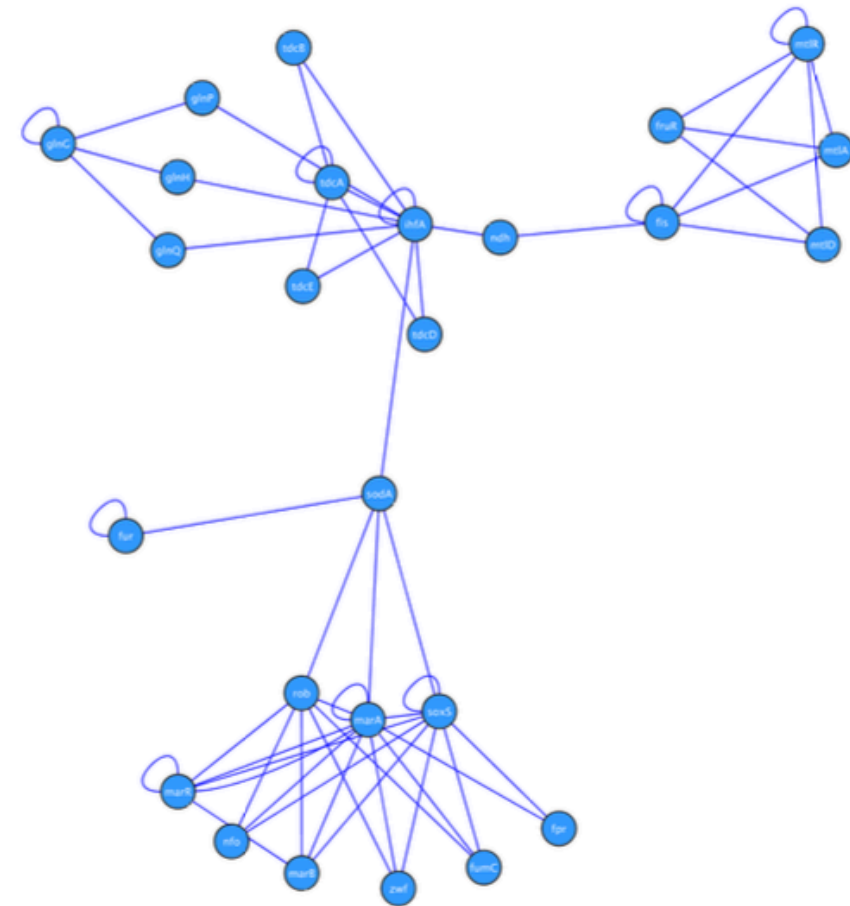
Undirected

- No distinction between source and target

- Co-expression (indirect)

- Binding (direct)

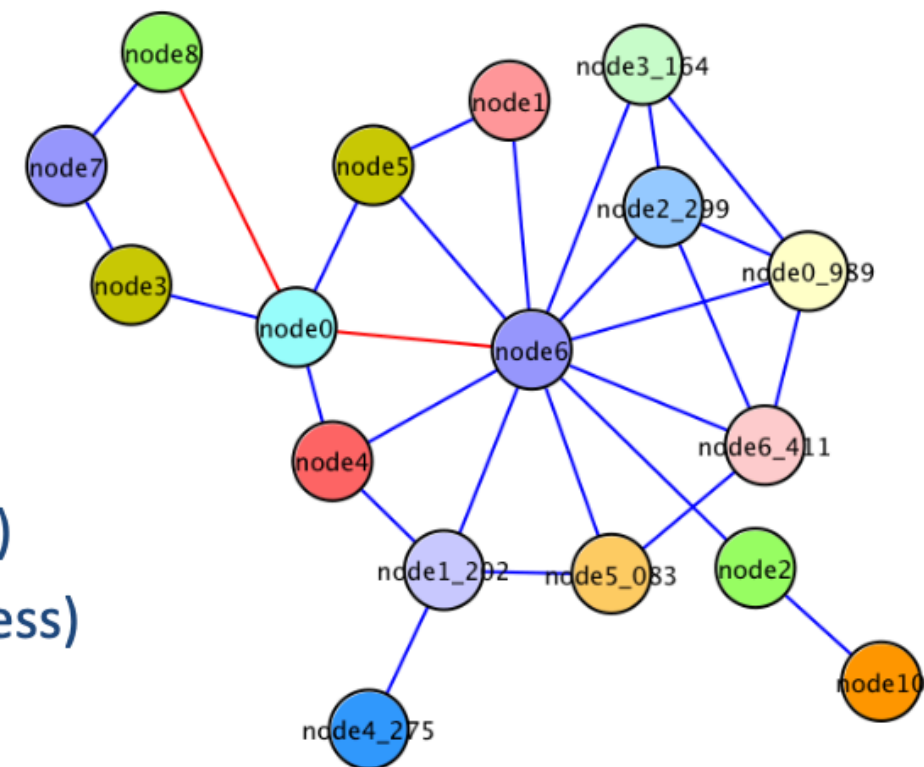
- Similarity/strength





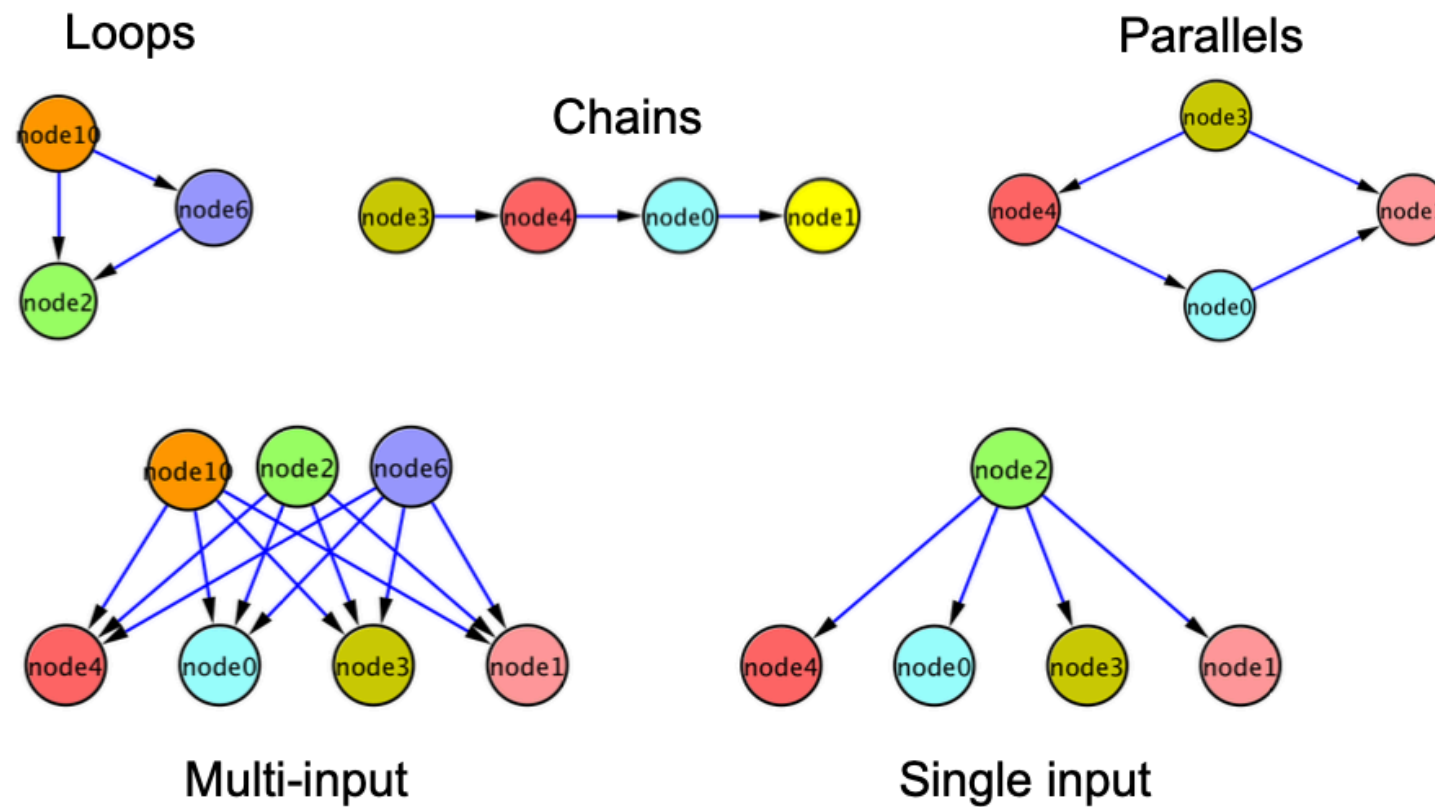
# Biological Networks - Basic Features

- Degree
  - Number of connections that a node has
- Distance
  - Number of connections between two nodes, in a shortest path
- Path
  - A sequence of connections
  - Is there a path (reachability)
  - Mean Shortest Path distance (closeness)
  - In how many shortest paths (betweenness)

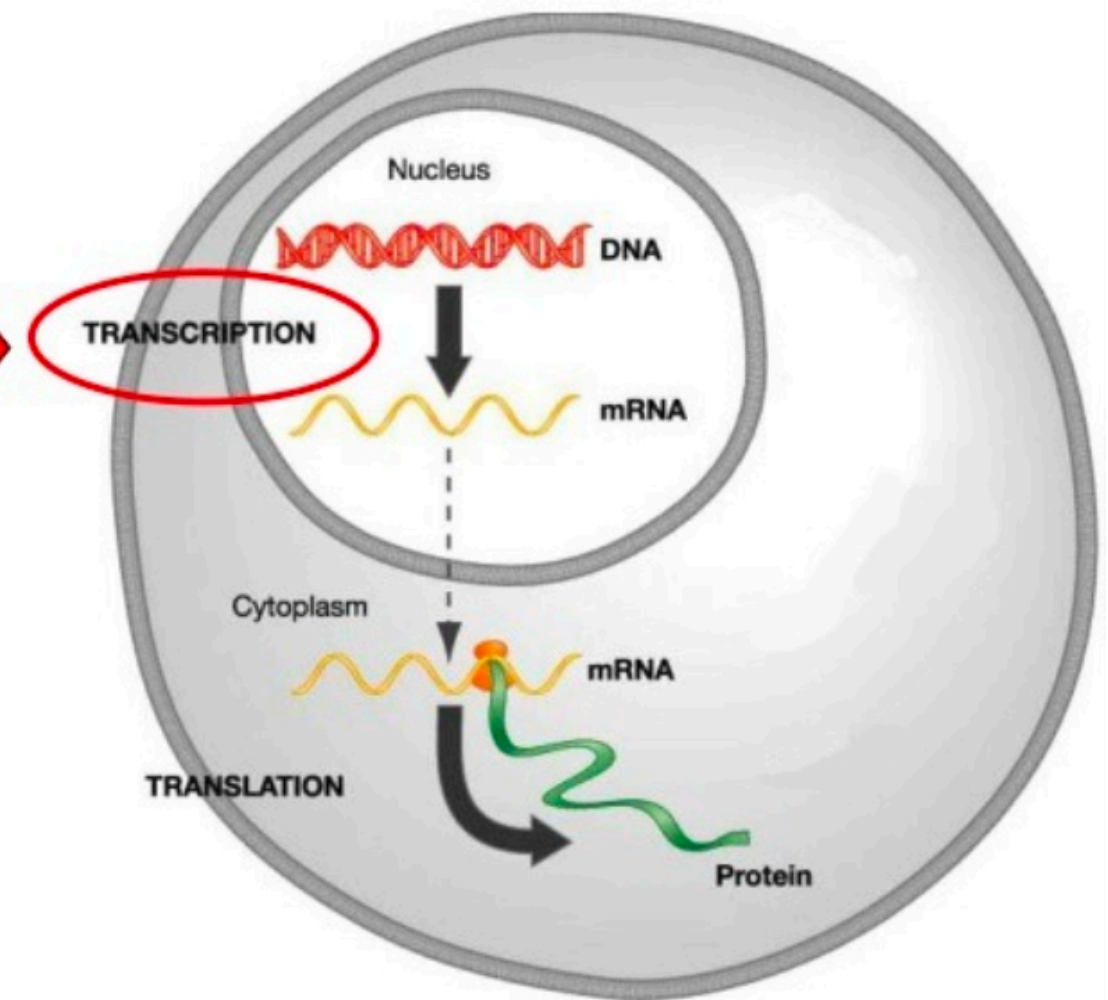
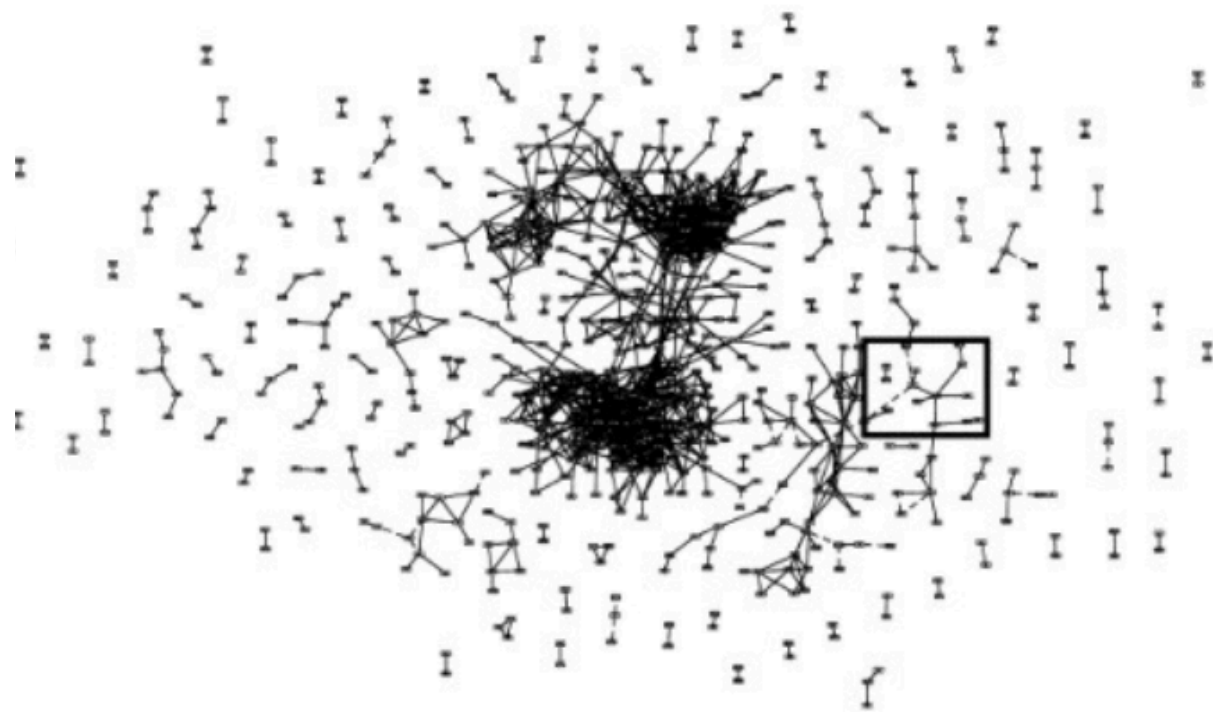


# Biological Networks - Basic Features

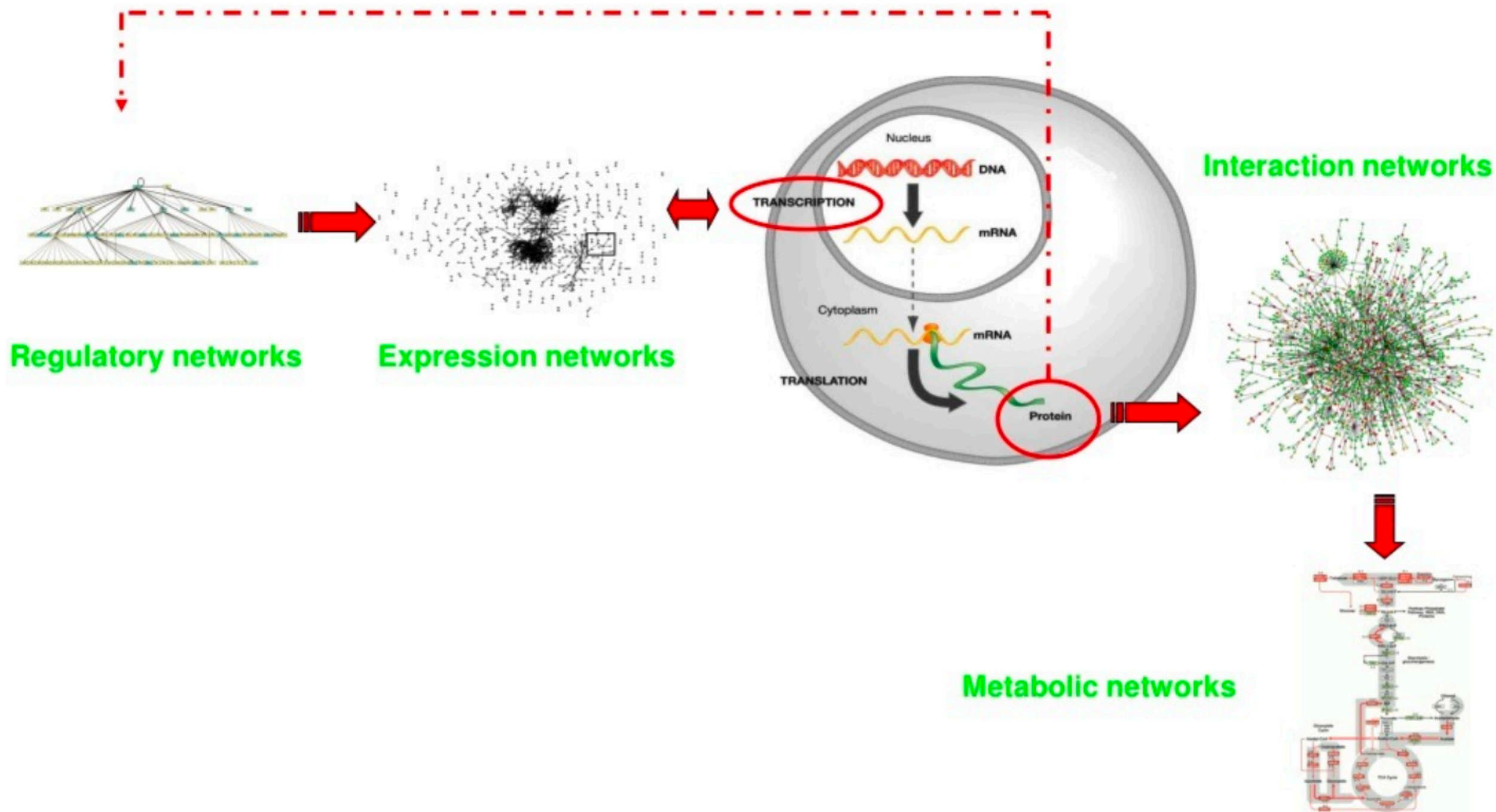
- Size of a network (Number of nodes)
- Density of a network (Proportion of the connections)
- Motifs/Cliques/Clusters/Sub-networks



# Expression networks



[Qian, et al, J. Mol. Bio., 314:1053-1066]

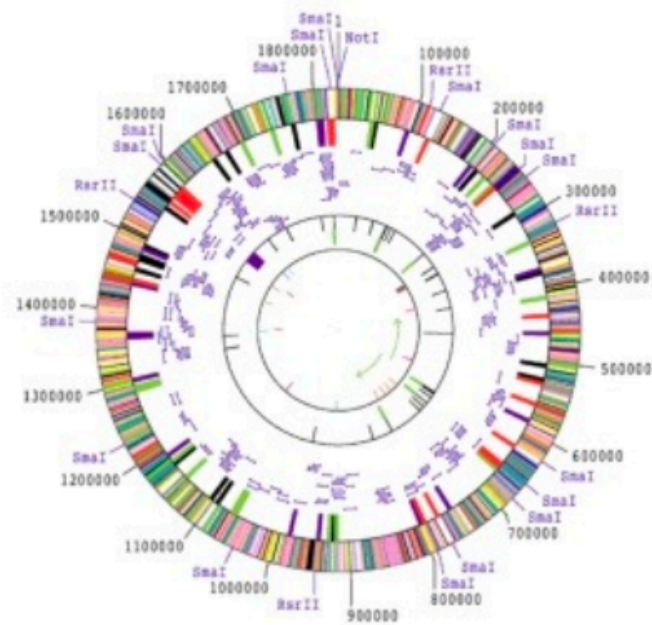


# Types of Biological Networks

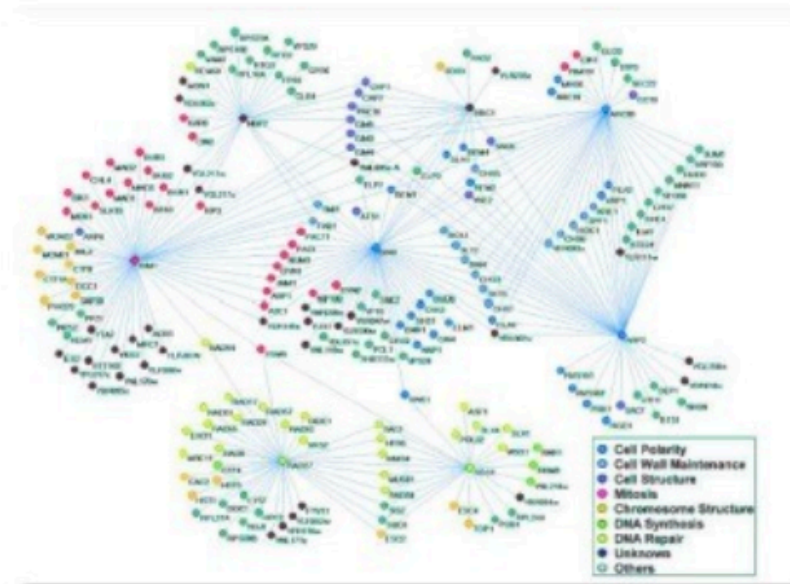
- DNA-Protein
  - Transcriptional regulatory networks
  - Methylation networks
- RNA-RNA
  - miRNA regulatory networks
- RNA-Protein
  - Splicing regulatory networks
- Protein-Protein
  - Co-expression networks
  - Co-localization networks
  - Co-evolution networks
  - Structure networks
  - Pathway networks
  - Protease regulatory networks
  - Signal transduction networks
  - Gene Ontology networks



## ... more biological networks



Gene order networks



Genetic interaction networks

[Boone]



Neural networks

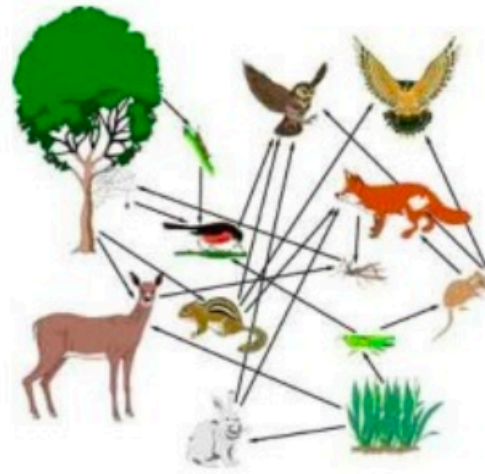
[Cajal]



# Other types of networks



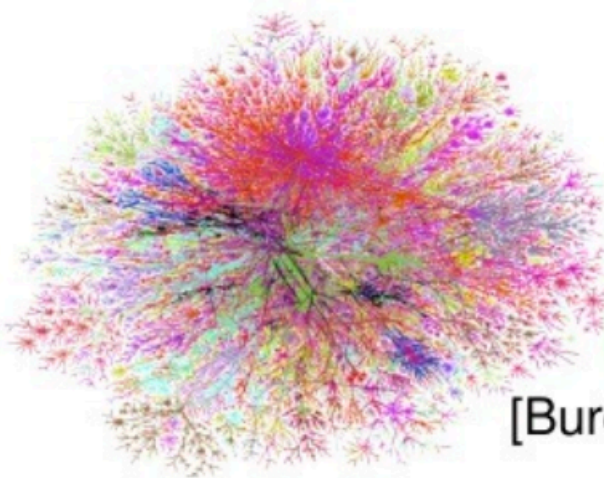
Disease  
Spread  
[Krebs]



Food Web



Electronic  
Circuit



Internet  
[Burch & Cheswick]



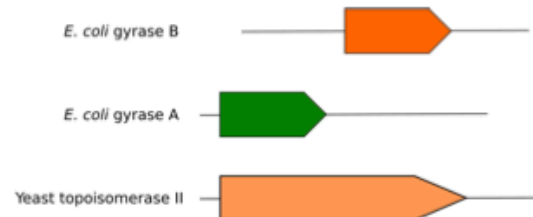
Social Network

# How to Build Biological Networks ?

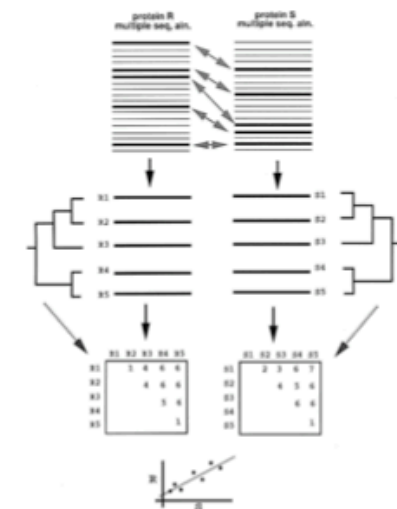
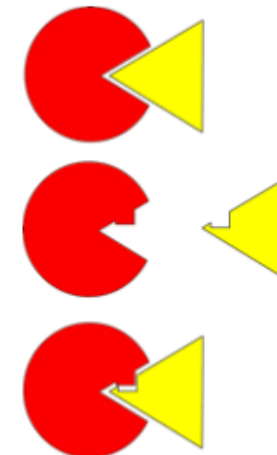
- Search/Retrieve from knowledge bases
- Predict from genome sequences
- Predict from “omics” data
- Predict from literature
- Integrate and analyze
- Meta-networks from genome scale data analysis

# Prediction from genome sequences

- Gene neighbor (gene cluster, gene order)
- Gene fusion (Rosetta stone)
- Phylogenetic profiling
- Co-evolution
- Mirror tree

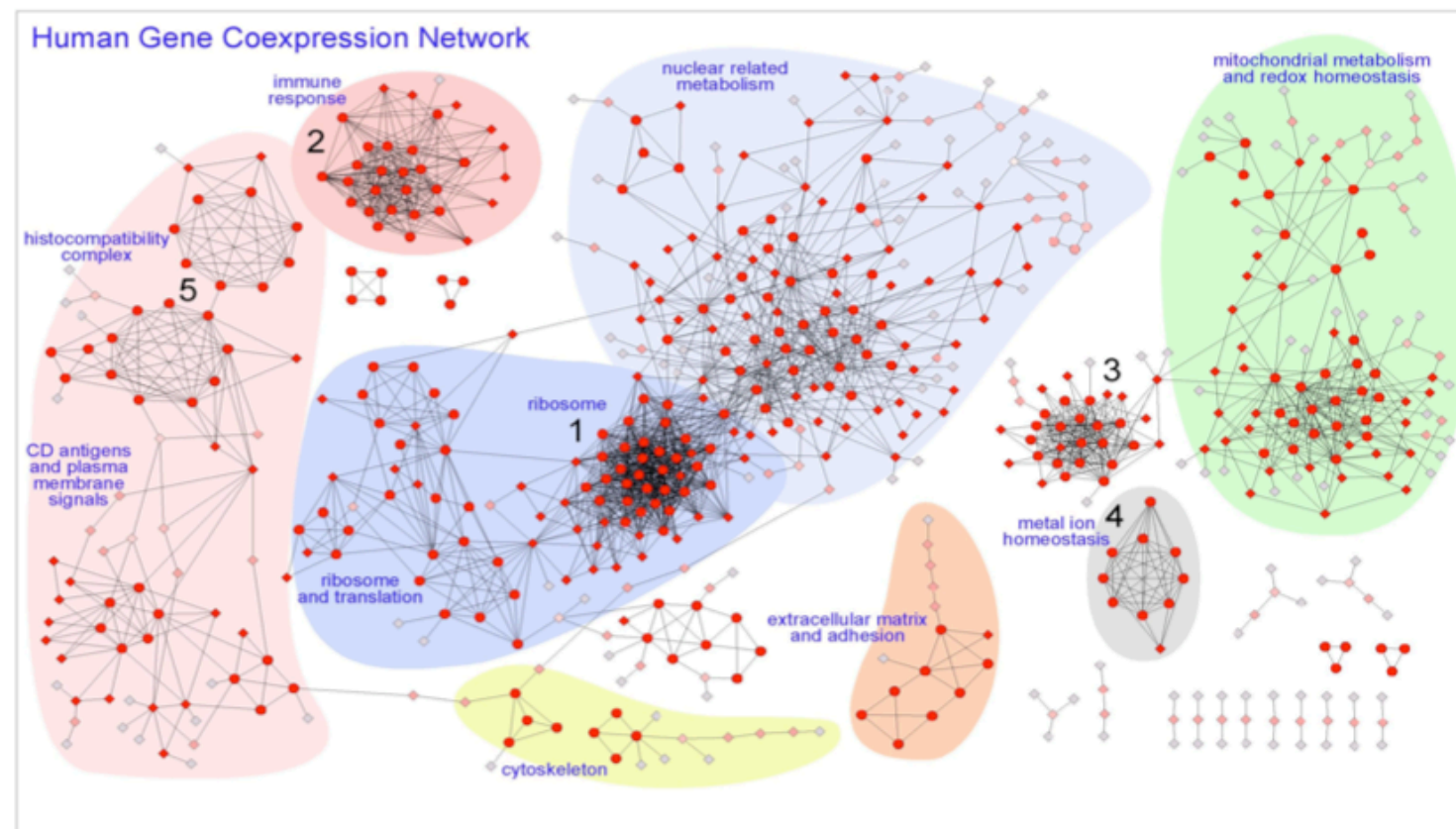


| Proteins | Genomes |    |    |
|----------|---------|----|----|
|          | EC      | HI | BS |
| P1       | 0       | 1  | 1  |
| P2       | 0       | 0  | 1  |
| P3       | 1       | 0  | 0  |
| P4       | 0       | 1  | 1  |

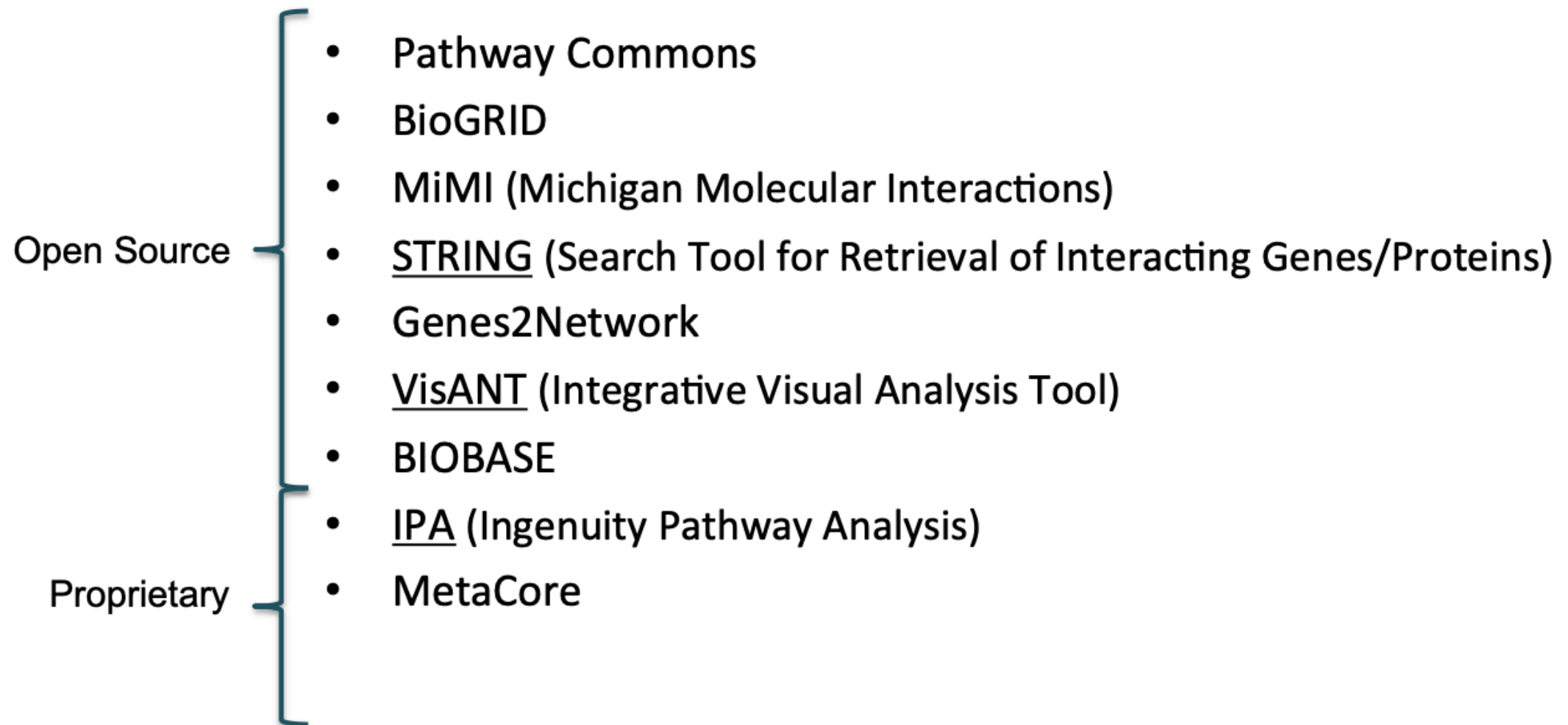


# Prediction from “omics” data

- **Co-expression** (Correlation, Mutual Information)
- **Integrated** (Classification)
- **Literature mining** (Natural Language Processing)



## Integrated Data Sources





# Outline

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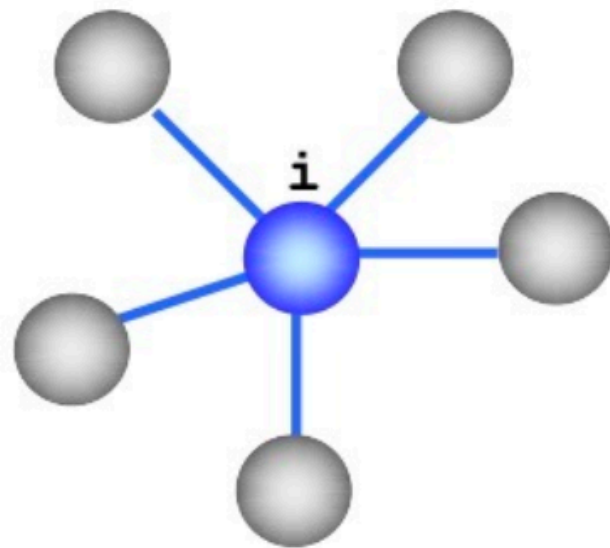
- Biological network
- **Network statistics**
- Network models
  - Network generation models
  - Scale free
  - Small world

| Indexes                                             | Formula                                                | Explanation                                                                                                             | Note                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Part I: network indexes for individual nodes</b> |                                                        |                                                                                                                         |                                                                                                                                                                                                                                                                                           |
| Connectivity                                        | $k_i = \sum_{j \neq i} a_{ij}$                         | $a_{ij}$ is the connection strength between nodes $i$ and $j$ .                                                         | It is also called node degree. It is the most commonly used concept for describing the topological property of a node in a network.                                                                                                                                                       |
| Stress centrality                                   | $SC_i = \sum_{jk} \sigma(j, i, k)$                     | $\sigma(j, i, k)$ is the number of shortest paths between nodes $j$ and $k$ that pass through node $i$ .                | It is used to describe the number of geodesic paths that pass through the $i^{\text{th}}$ node. High Stress node can serve as a broker.                                                                                                                                                   |
| Betweenness                                         | $B_i = \sum_{jk} \frac{\sigma(j, i, k)}{\sigma(j, k)}$ | $\sigma(j, k)$ is the total number of shortest paths between $j$ and $k$ .                                              | It is used to describe the ratio of paths that pass through the $i^{\text{th}}$ node. High Betweenness node can serve as a broker similar to stress centrality.                                                                                                                           |
| Eigenvector centrality                              | $EC_i = \frac{1}{\lambda} \sum_{j \in M(i)} EC_j$      | $M(i)$ is the set of nodes that are connected to the $i^{\text{th}}$ node and $\lambda$ is a constant eigenvalue.       | It is used to describe the degree of a central node that it is connected to other central nodes.                                                                                                                                                                                          |
| Clustering coefficient                              | $CC_i = \frac{2l_i}{k_i'(k_i' - 1)}$                   | $l_i$ is the number of links between neighbors of node $i$ and $k_i'$ is the number of neighbors of node $i$ .          | It describes how well a node is connected with its neighbors. If it is fully connected to its neighbors, the clustering coefficient is 1. A value close to 0 means that there are hardly any connections with its neighbors. It was used to describe hierarchical properties of networks. |
| Vulnerability                                       | $V_i = \frac{E - E_i}{E}$                              | $E$ is the global efficiency and $E_i$ is the global efficiency after the removal of the node $i$ and its entire links. | It measures the decrease of node $i$ on the system performance if node $i$ and all associated links are removed.                                                                                                                                                                          |

## Part II: The overall network topological indexes

|                                     |                                                         |                                                                                                                                                                                                            |                                                                                                                                                       |
|-------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average connectivity                | $avgK = \frac{\sum_{i=1}^n k_i}{n}$                     | $k_i$ is degree of node $i$ and $n$ is the number of nodes.                                                                                                                                                | Higher $avgK$ means a more complex network.                                                                                                           |
| Average geodesic distance           | $GD = \frac{1}{n(n-1)} \sum_{i \neq j} d_{ij}$          | $d_{ij}$ is the shortest path between node $i$ and $j$ .                                                                                                                                                   | A smaller $GD$ means all the nodes in the network are closer.                                                                                         |
| Geodesic efficiency                 | $E = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{1}{d_{ij}}$ | all parameters shown above.                                                                                                                                                                                | It is the opposite of $GD$ . A higher $E$ means that the nodes are closer.                                                                            |
| Harmonic geodesic distance          | $HD = \frac{1}{E}$                                      | $E$ is geodesic efficiency.                                                                                                                                                                                | The reciprocal of $E$ , which is similar to $GD$ but more appropriate for disjoint graph.                                                             |
| Centralization of degree            | $CD = \sum_{i=1}^n (\max(k) - k_i)$                     | $\max(k)$ is the maximal value of all connectivity values and $k_i$ represents the connectivity of $i^{\text{th}}$ node. Finally this value is normalized by the theoretical maximum centralization score. | It is close to 1 for a network with star topology and in contrast close to 0 for a network where each node has the same connectivity.                 |
| Centralization of betweenness       | $CB = \sum_{i=1}^n (\max(B) - B_i)$                     | $\max(B)$ is the maximal value of all betweenness values and $B_i$ represents the betweenness of $i^{\text{th}}$ node. Finally this value is normalized by the theoretical maximum centralization score.   | It is close to 0 for a network where each node has the same betweenness, and the bigger the more difference among all betweenness values.             |
| Centralization of stress centrality | $CS = \sum_{i=1}^n (\max(SC) - SC_i)$                   | $\max(SC)$ is the maximal value of all stress centrality values and $SC_i$ represents the stress centrality of $i^{\text{th}}$ node. Finally this value is normalized by the                               | It is close to 0 for a network where each node has the same stress centrality, and the bigger the more difference among all stress centrality values. |

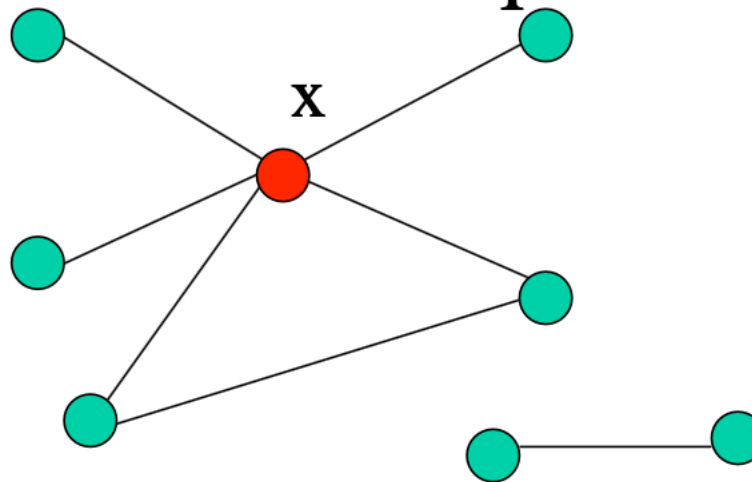
- **Degree of a node:** the number of edges incident on the node



Degree of node  $i = 5$

# Betweenness

- The total number of shortest paths that pass through a vertex  $x$  (normalized by #pairs in the connected component containing  $x$ ).
- Indicates if vertex is important for the traffic

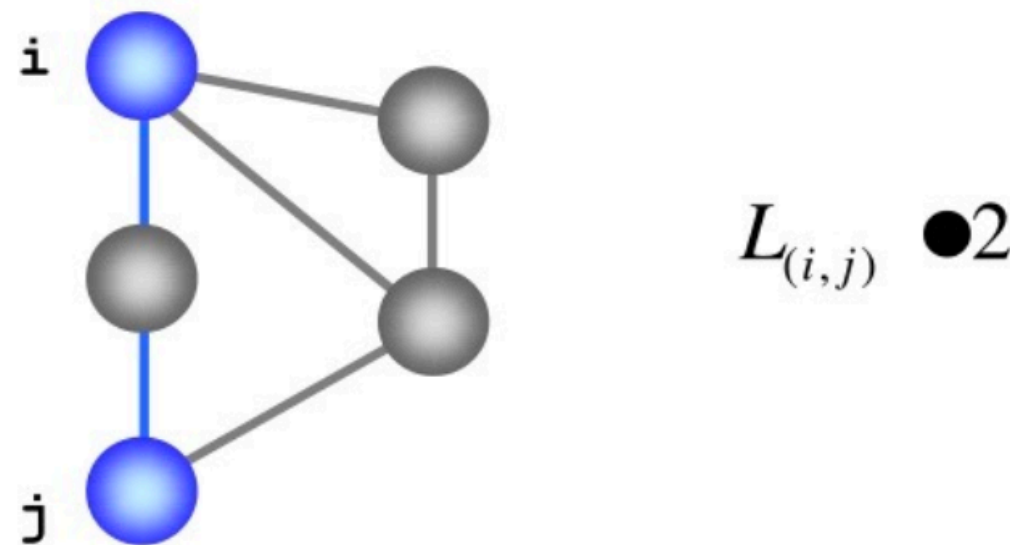


- Also called betweenness centrality



# Characteristic path length $\square$ GLOBAL property

- $L_{(i,j)}$  is the number of edges in the shortest path between vertices  $i$  and  $j$



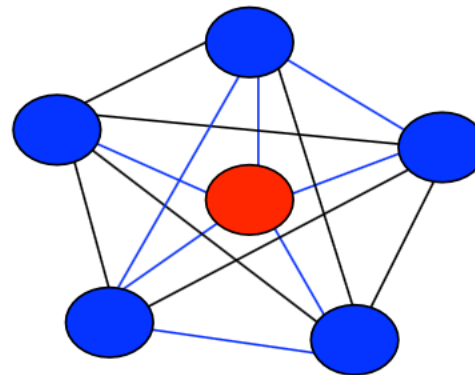
- The characteristic path length  $L$  of a graph is the average of the  $L_{(i,j)}$  for every possible pair  $(i,j)$

Networks with small values of  $L$  are said to have the “small world property”

# Connectivity/clustering coef.

- Characterizes “density” of connections
- For each node, consider its neighbors and see what percentage of possible connections between them are realized

$z = \text{\#neighbors ;}$   
 $y = \text{\#connections}$   
between the  
neighbors  
 $C = 2y / z(z-1)$



$\bar{C}$  = average connectivity = prob. there exists a connection between two neighbors

For ER model  $\bar{C} = \bar{k} / N$

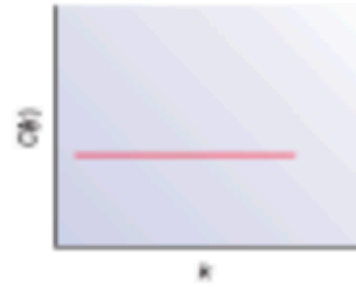
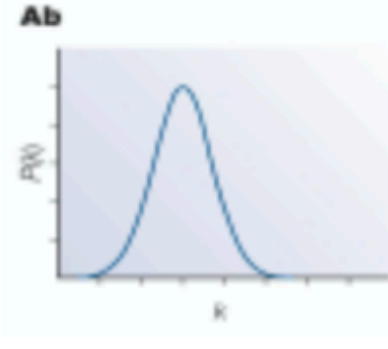
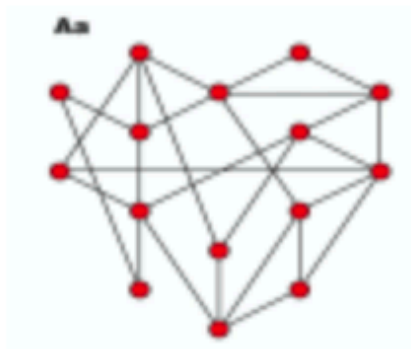
# Outline

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- Biological network
- Network statistics
- **Network models**
  - Network generation models
  - Scale free
  - Small world

# Various types of networks

## A. Random Networks [Erdos and Rényi (1959, 1960)]

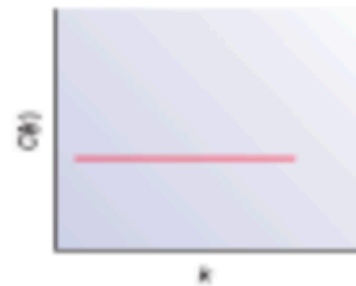
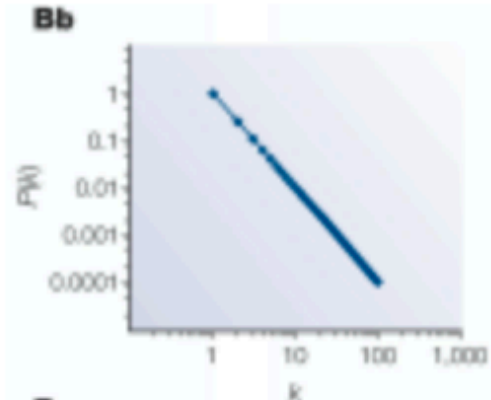
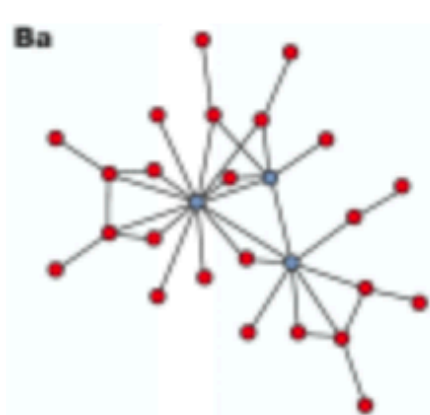


*Mean path length  $\sim \ln(k)$*

*Phase transition:*

*Connected if:*

## B. Scale Free [Price,1965 & Barabasi,1999]

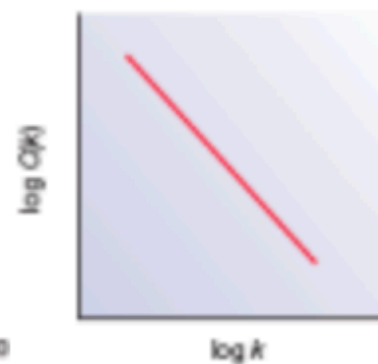
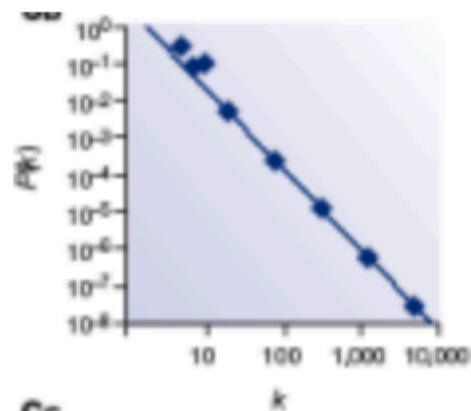
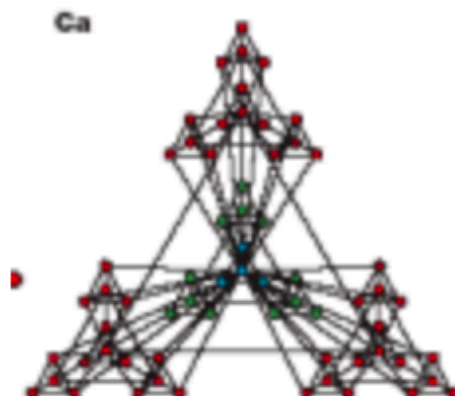


*Mean path length  $\sim \ln \ln(k)$*

Preferential attachment. Add proportionally to connectedness

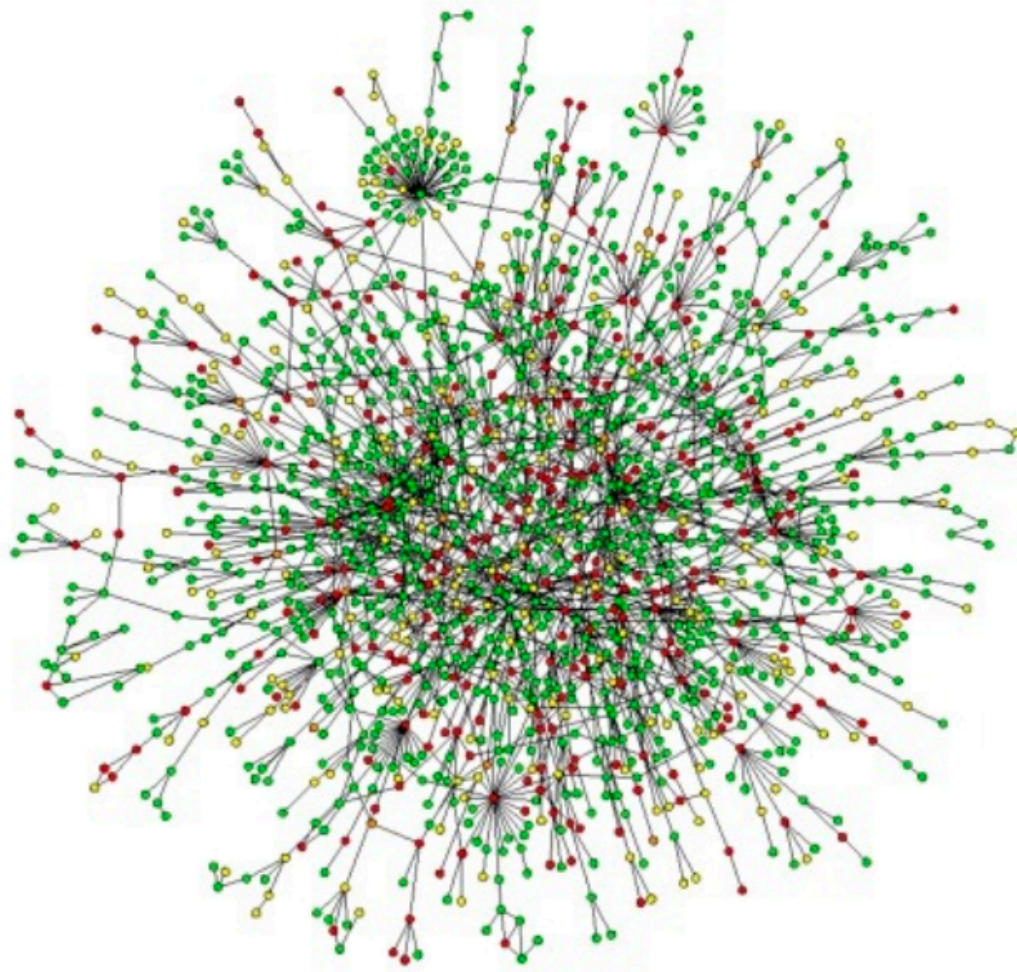


## C. Hierarchical



Copy smaller graphs and let them keep their connections.

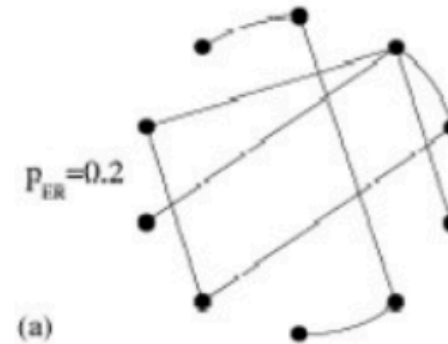
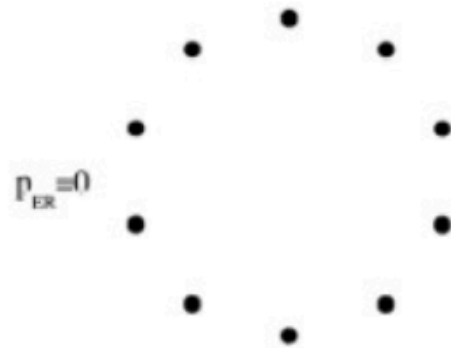
# Network generation models



- Erdos-Renyi (1960)
- Watts-Strogatz (1998)
- Barabasi-Albert (1999)



# The Erdős-Rényi [ER] model (1960)



- Start with  $N$  vertices and no edges
- Connect each pair of vertices with probability  $P_{ER}$

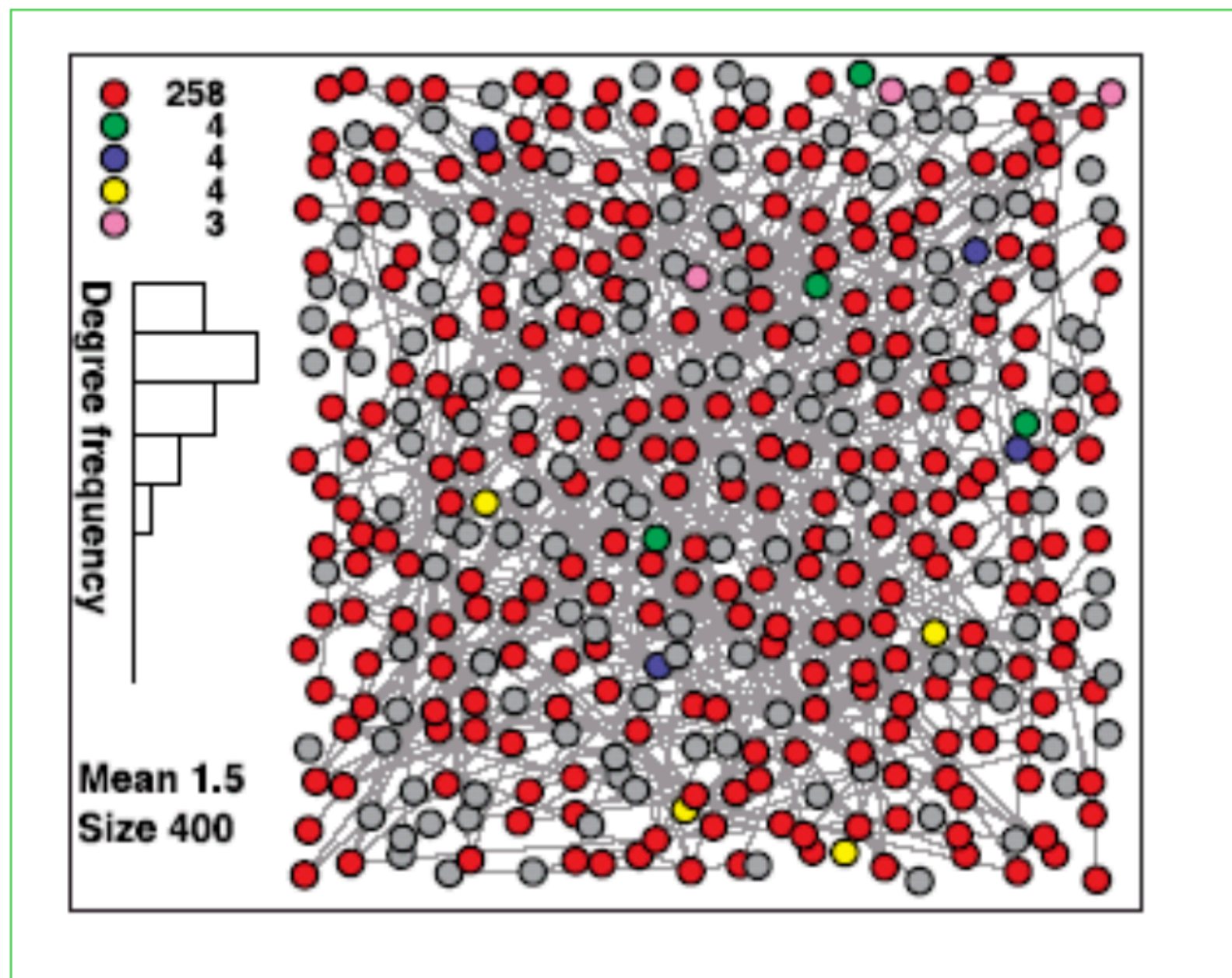
**Important result:** many properties in these graphs appear quite suddenly, at a threshold value of  $P_{ER}(N)$

- If  $P_{ER} \sim c/N$  with  $c < 1$ , then almost all vertices belong to isolated trees
- Cycles of all orders appear at  $P_{ER} \sim 1/N$

## Model 1: Erdos-Renyi model

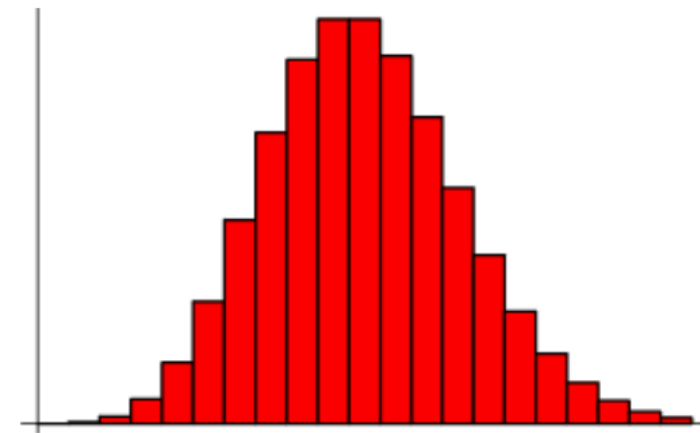
- Erdos-Renyi model:
  - With probability  $p$  put an edge between any pair of vertices.

<http://www.ams.org/notices/200404/comm-networks.pdf>



# The degree distribution in Erdos-Renyi model: **Poisson**

$$p(k) = \frac{e^{-\bar{k}} \bar{k}^k}{k!}$$



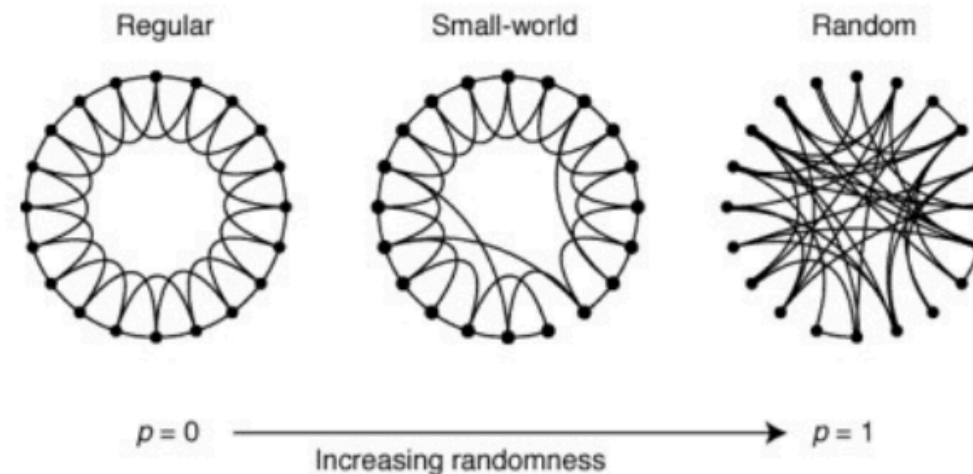
$P(k)$  – probability of node having degree  $k$

$\bar{k} = \sum_k k p(k)$  (average degree)

This distribution is approached in ER model as  $\bar{k}$  goes to infinity under assumption that  $k$  is fixed



# The Watts-Strogatz [WS] model (1998)



- Start with a regular network with  $N$  vertices
- Rewire each edge with probability  $p$

For  $p=0$  (Regular Networks):

- high clustering coefficient
- high characteristic path length

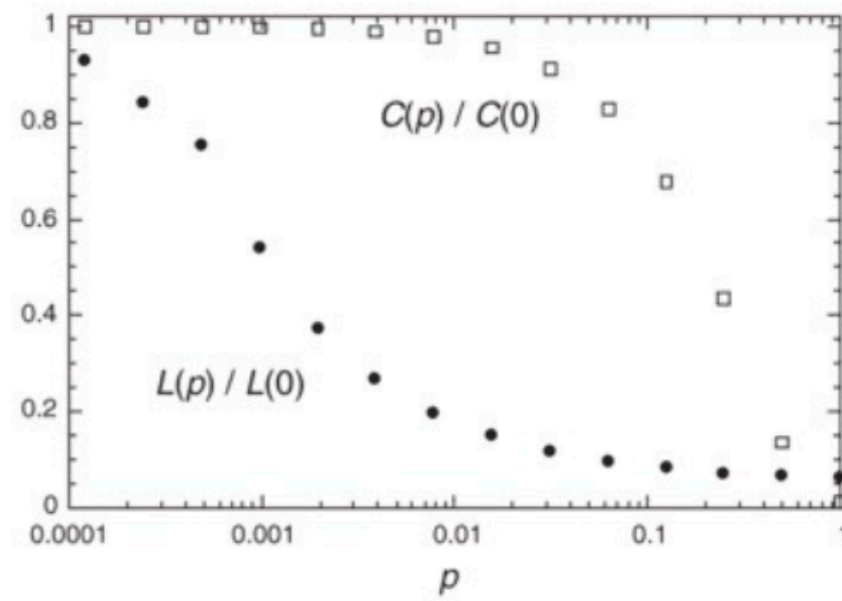
For  $p=1$  (Random Networks):

- low clustering coefficient
- low characteristic path length

QUESTION: What happens for intermediate values of  $p$ ?



1) There is a broad interval of  $p$  for which  $L$  is small but  $C$  remains large



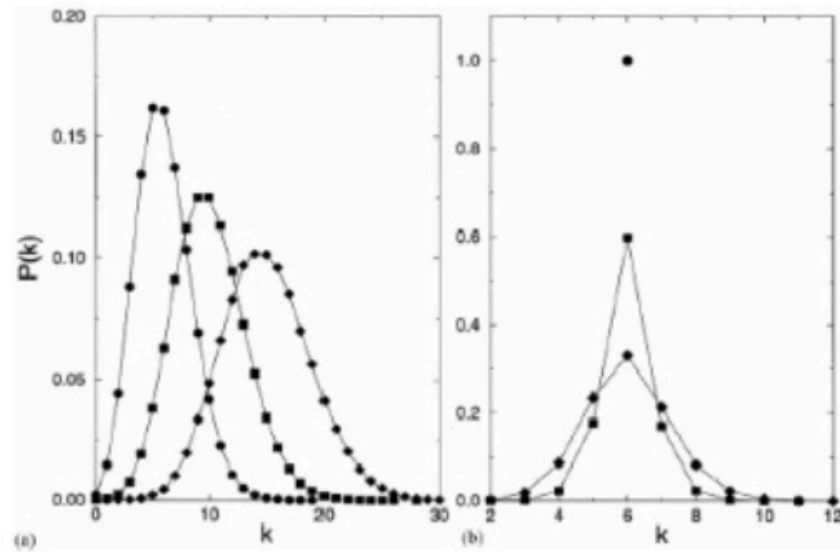
2) Small world networks are common :

**Table 1 Empirical examples of small-world networks**

|                   | $L_{\text{actual}}$ | $L_{\text{random}}$ | $C_{\text{actual}}$ | $C_{\text{random}}$ |
|-------------------|---------------------|---------------------|---------------------|---------------------|
| Film actors       | 3.65                | 2.99                | 0.79                | 0.00027             |
| Power grid        | 18.7                | 12.4                | 0.080               | 0.005               |
| <i>C. elegans</i> | 2.65                | 2.25                | 0.28                | 0.05                |

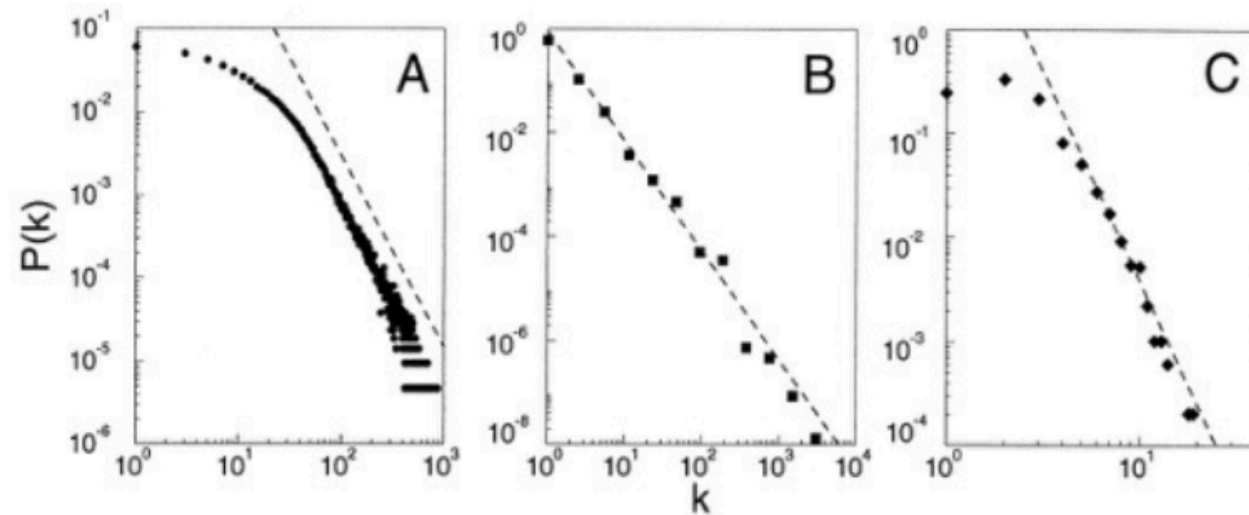
# The Barabási-Albert [BA] model (1999)

Look at the distribution of degrees



ER Model

WS Model



actors

power grid

www

The probability of finding a highly connected node decreases exponentially with  $k$

$$P(K) \sim K^{-2}$$

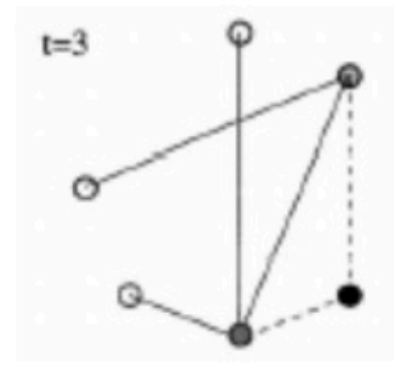
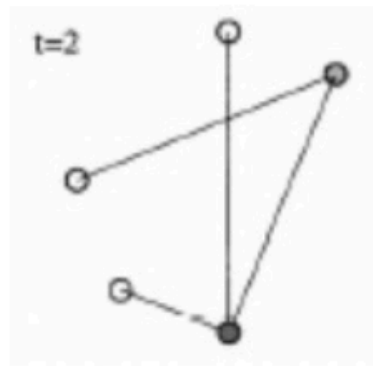
- two problems with the previous models:

1.  $N$  does not vary
2. the probability that two vertices are connected is uniform

- GROWTH: starting with a small number of vertices  $m_0$  at every timestep add a new vertex with  $m \leq m_0$

- PREFERENTIAL ATTACHMENT: the probability  $\Pi$  that a new vertex will be connected to vertex  $i$  depends on the connectivity of that vertex:

$$\Pi(k_i) \propto \frac{k_i}{\sum_j k_j}$$

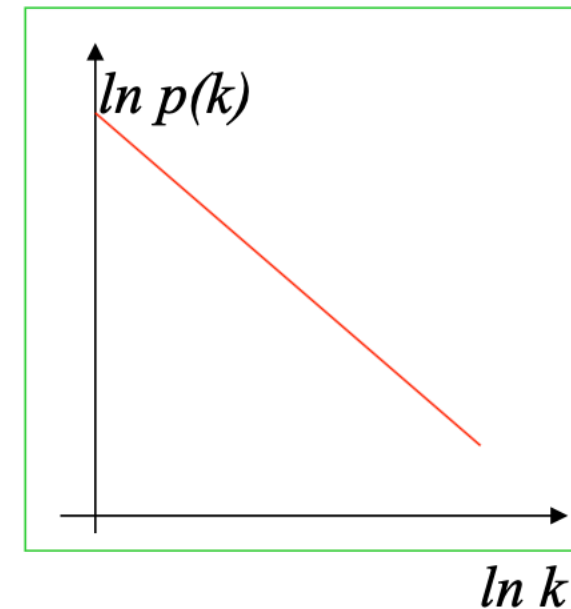


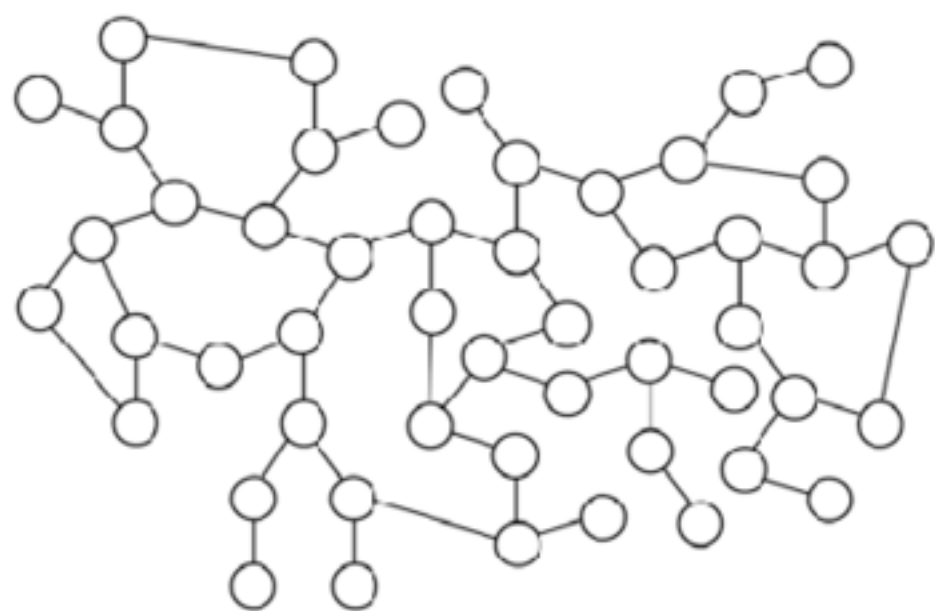
## Scale free

Vertex degree distribution for biological (and many other “real world” networks) better approximated by is Scale free distribution

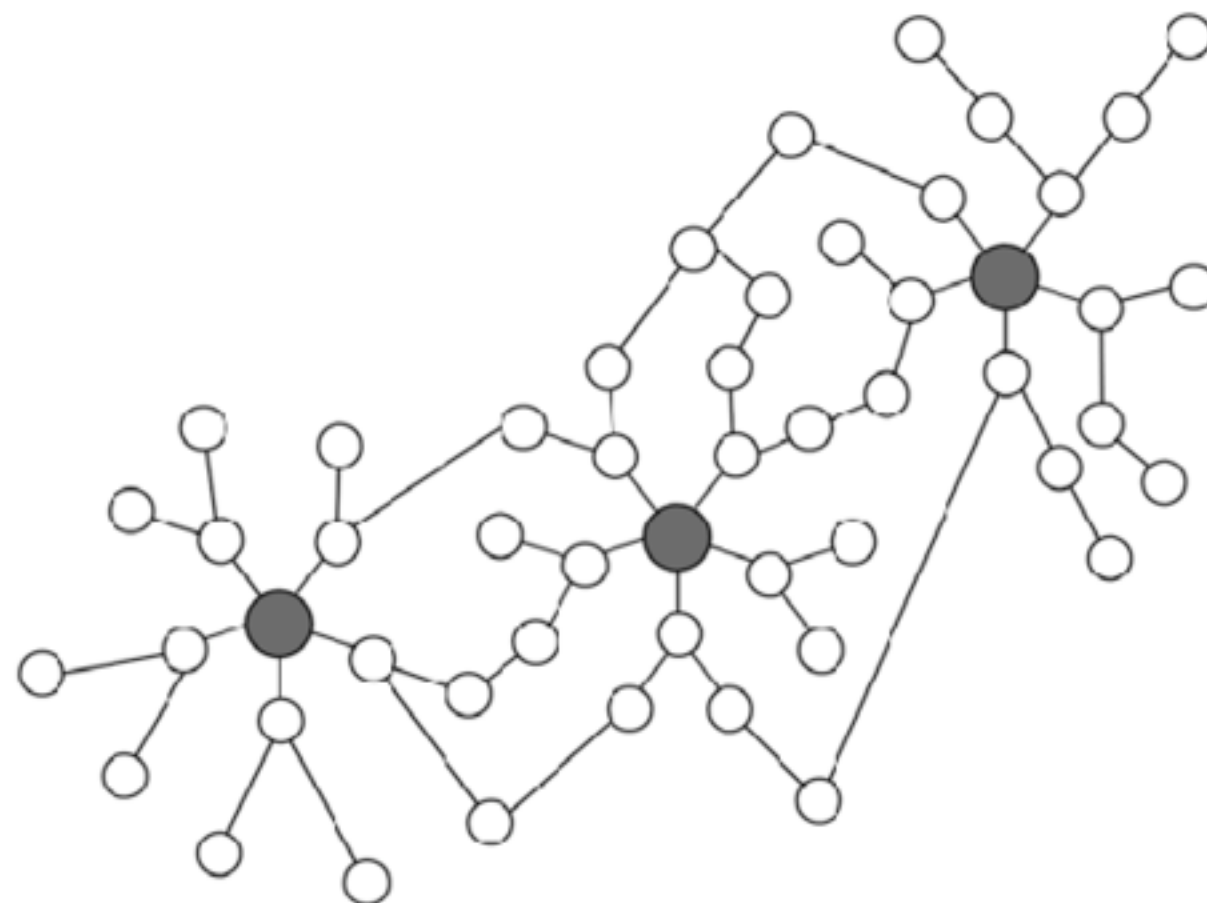
$$P(k) \sim k^{-\gamma}$$

$P(k)$  probability of node of degree  $k$



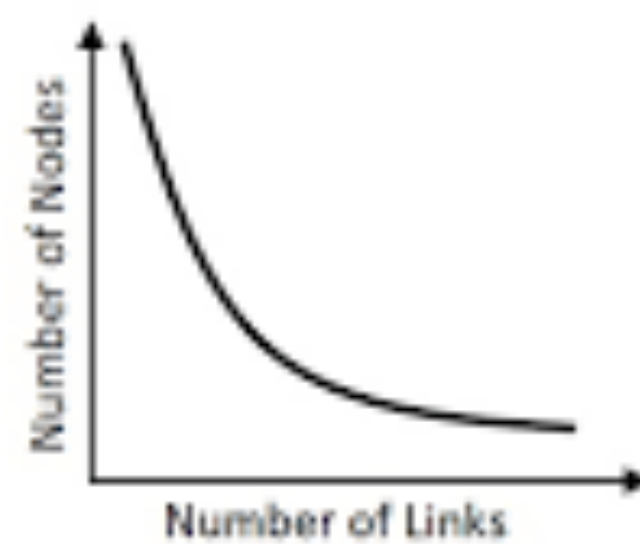
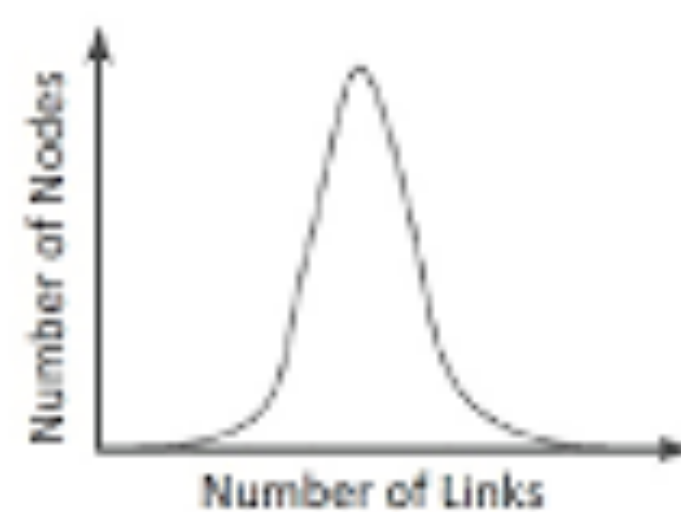


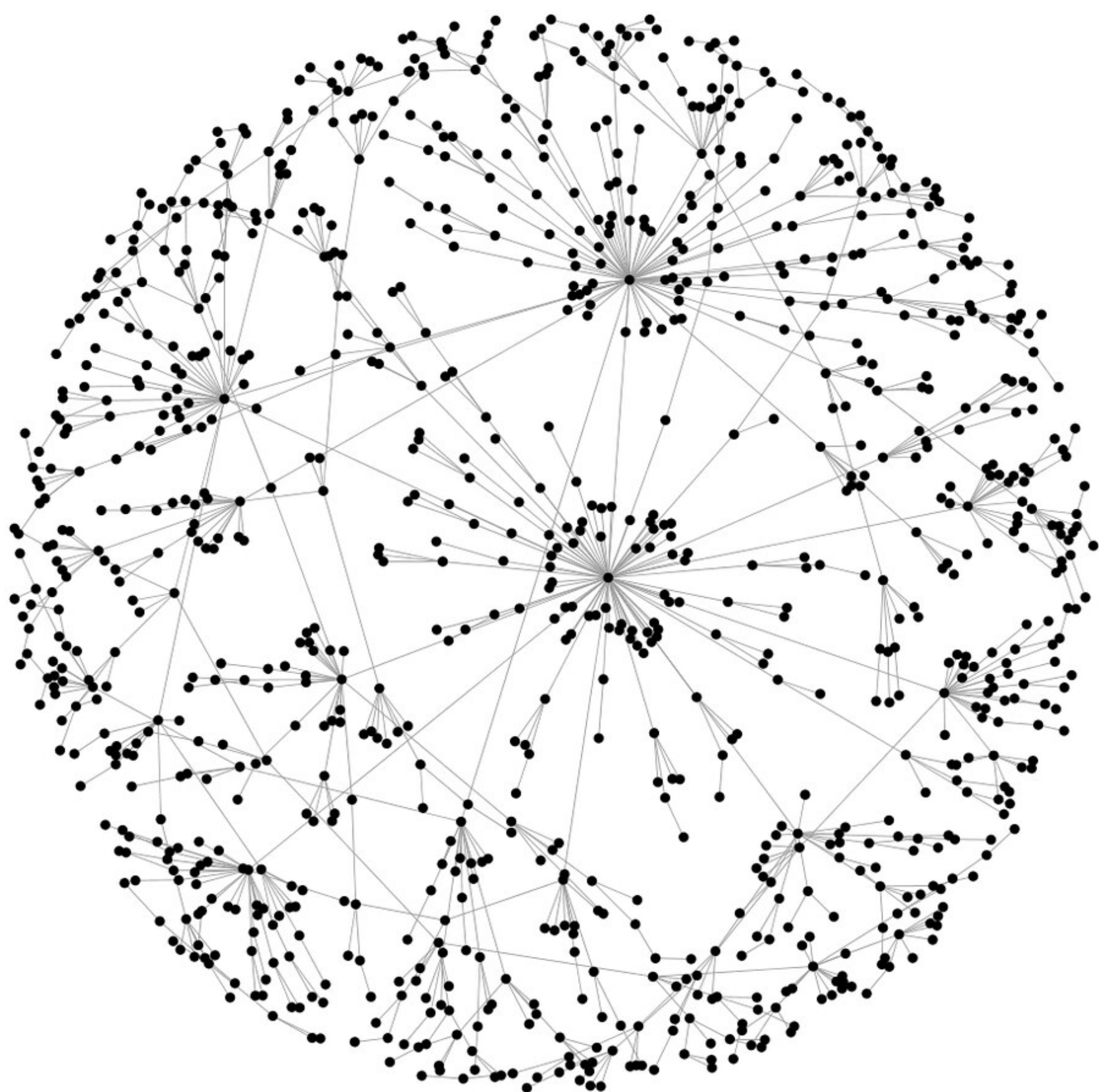
**(A) Random network**



**(B) Scale-free network**









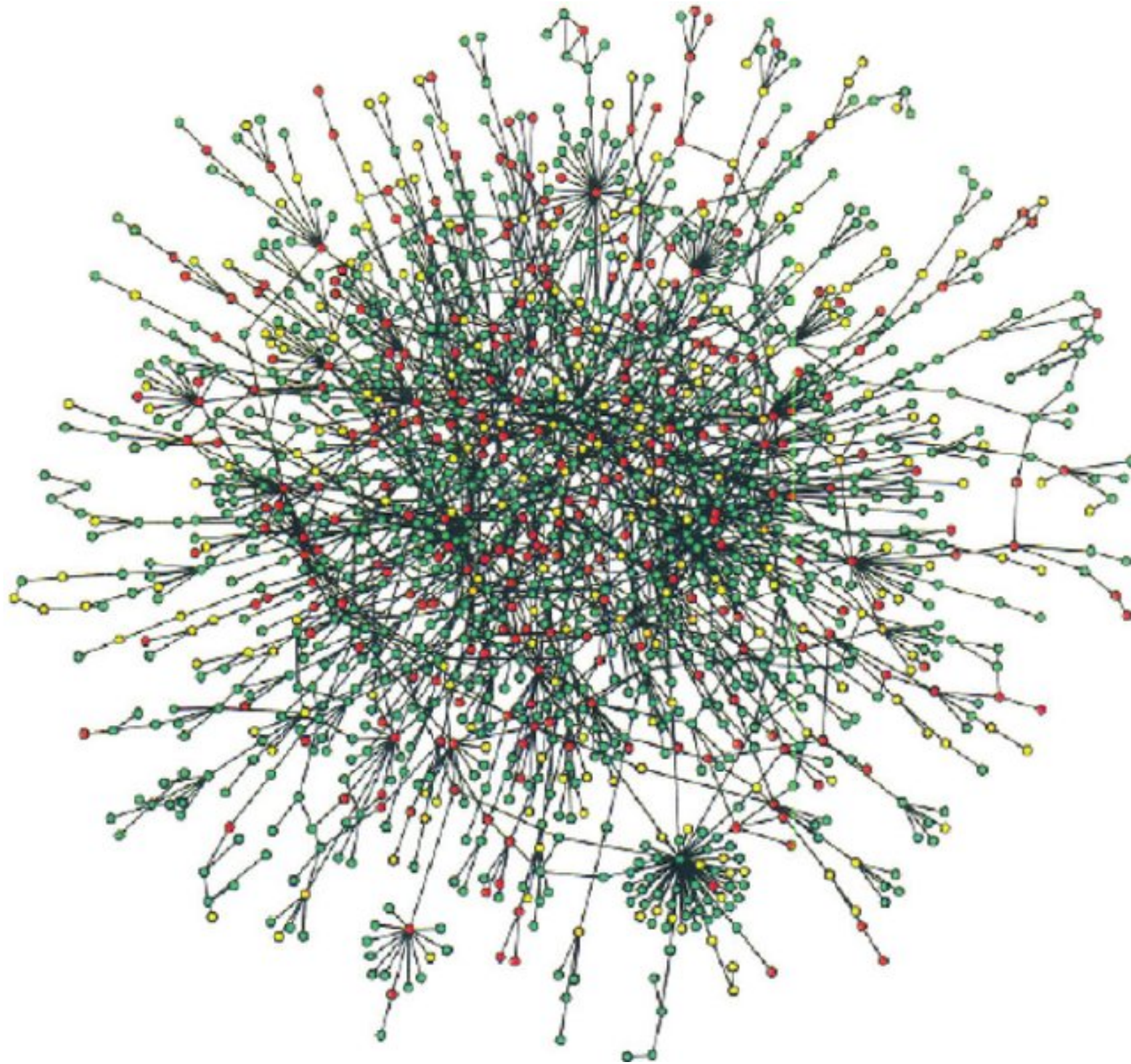
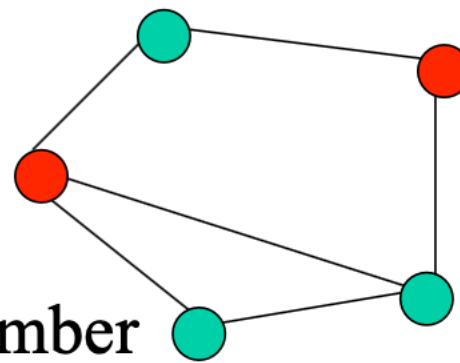


Figure 1. Example scale-free network, representing interacting proteins in yeast (Barabási and Bonabeau 2003).

# Small worlds

- **Distance** between two nodes in a net = the **smallest number of steps** one can take to reach one node from the other



The distance between red nodes = 2

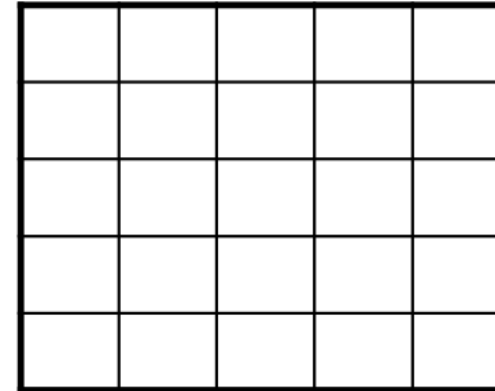
Example: Erdos number

$l_{\max}$  = longest shortest path = diameter of the network

# Small worlds- cont.

$p(l)$  = the probability that the shortest path between two random nodes is  $l$

$$\bar{l} = \sum_l l p(l)$$



*In a grid  $\bar{l} \sim \sqrt{N}$*

*In a tree  $\bar{l} \sim \ln N / \ln \bar{z}$  where  $\bar{z}$  – average degree*

*The fact that  $\bar{l}$  tends to be small (relative to a regular grid) is called small world effect.*

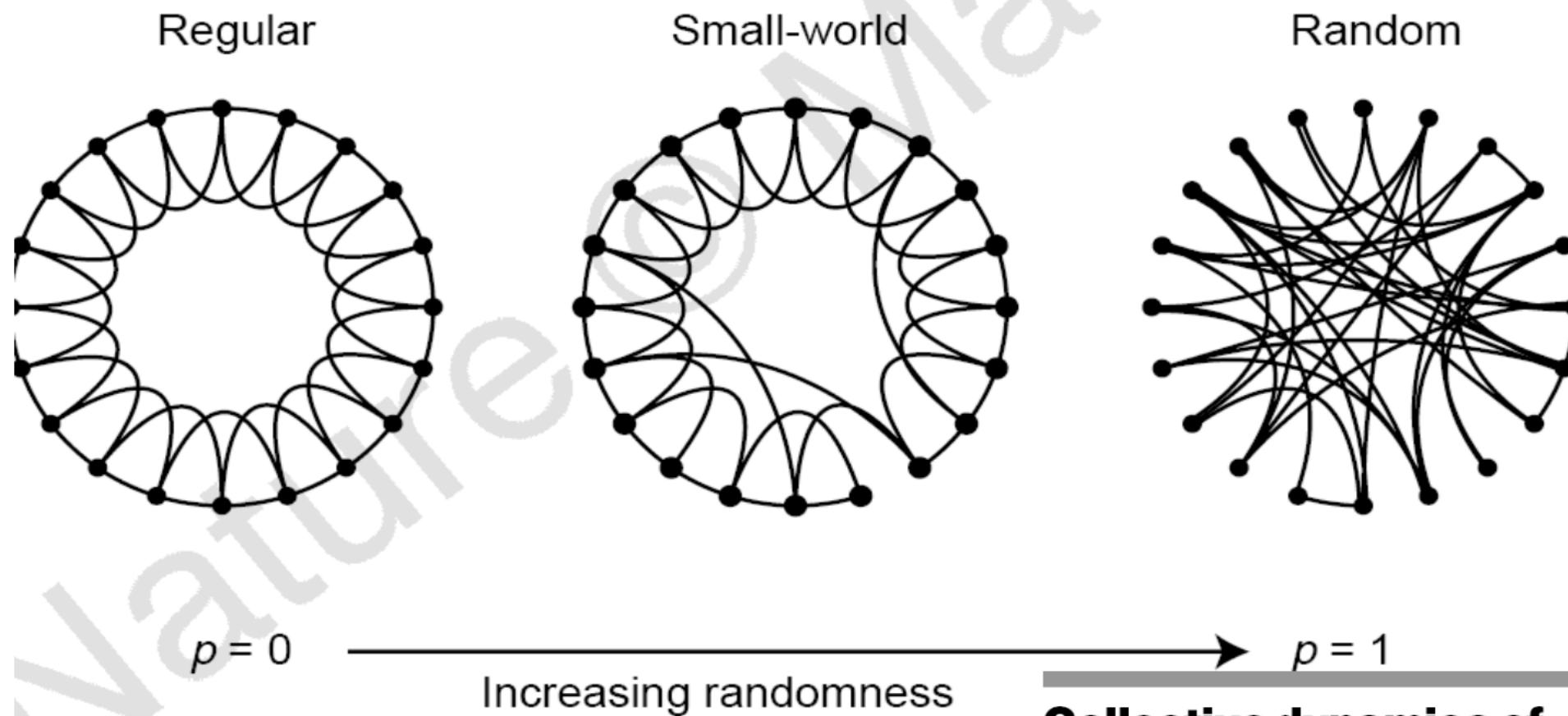
*Small world:*

$$C(\text{small world}) \gg C(\text{random})$$

$$L(\text{small world}) \sim L(\text{random})$$



*Connecting randomly chosen vertices of regular lattice converts it to a small world net.*



### **Collective dynamics of 'small-world' networks**

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